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Hybrid data balancing and ensemble learning with SHAP-based explainability to predict cardiovascular disease using a unified dataset

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Abstract

Objectives: Heart disease is currently one of the most serious diseases, contributing to a large share of patient deaths. Diagnosing it is a challenging procedure that needs to be done both quickly and accurately. **Methods:** This research introduces a practical machine learning framework aimed at improving heart disease prediction using multiple datasets. To build a more reliable model, three separate datasets were first combined into a single, enriched source of data. Since the data was imbalanced, a hybrid balancing technique using the Instance Hardness Threshold (IHT) and Tomek Links was applied to ensure fairer learning across classes. All preprocessing (scaling, balancing, hyperparameter tuning) was done exclusively within the training set of each fold in stratified 5-fold cross validation, with the hyperparameter tuning further embedded in an inner 3-fold cross validation. A total of 12 machine learning models were trained and compared, with the top-performing models ensembled using soft- and hard-voting. **Results:** The most successful model, a soft voting combination of XGBoost, Extra Trees, Random Forest, MLP and Decision Tree, outperformed the others with an accuracy of 96.02% (95% CI: 95.47 – 96.56%) with a precision of 97.83% and a recall of 94.37% using the fold-wise protocol designed to prevent preprocessing leakage. In addition to accuracy, calibration analysis and clinical decision-curve analysis showed that the model's predicted probabilities are suitable for a clinical context, and a sensitivity–specificity analysis revealed an optimal decision threshold of 0.323. Finally, SHAP was applied to clarify the impact of each feature; chest pain type and thalassemia status were found to be the most influential predictors. **Conclusion:** The proposed ensemble model, based on a hybrid-balanced SHAP interpretable model holds great promise for clinical applications and transparency in heart disease prediction.

Keywords

Machine learning, cardiovascular disease, data imbalance, ensemble learning, SHAP, data leakage, model calibration

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1. Introduction

Cardiovascular diseases (CVDs) have an elevated mortality rate worldwide, and most of the deaths are brought on by delayed diagnosis. More than 30% of fatalities worldwide are attributable to CVDs, which cause 17 million deaths annually,

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according to WHO estimates.[1] In India, 45% of fatalities among those aged 40 to 69 are attributable to CVD.[2] Part of this burden is due to delayed diagnosis; traditional diagnostic methods such as angiography, biomarker testing and electrocardiogram (ECG) analysis are accurate, but time-consuming and labour-intensive, and not always available early enough in a patient's treatment to avoid adverse outcomes. Adding machine learning (ML) algorithms that can recognize illness using pertinent datasets is an interesting way to address this. Research in Ref. 3 has looked into machine learning and deep learning techniques to increase the accuracy of heart disease prediction models. Motwani et al.[4] compared the accuracy of machine learning (ML) compared to clinical data or coronary computed tomographic angiography (CCTA) alone in predicting 5-year all-cause mortality in patients who underwent CCTA. Experimental research by[5,6] also indicates that ML can be utilized to aid in the early diagnosis. In the medical field, it is more important to make prediction on time and accurately rather than categorization incorrectly, so Azmi et al.[7] discussed some machine learning methods for categorization and forecasting algorithms. Similarly, Aslan et al.[8] demonstrated that targeted preprocessing and data augmentation techniques substantially improved deep learning-based arrhythmia detection accuracy, underscoring the broader value of careful data preparation in cardiac ML pipelines, a theme directly relevant to the preprocessing and balancing strategy adopted in this study.

Several researchers have used ML classifiers for predicting heart disease and have taken one of two paths: single-classifier approaches[9–14] or ensemble approaches that use multiple classifiers.[15,16] Ali et al.[9] analyzed risk factors and survival prediction in heart failure patients using ML, achieving the highest accuracy of 97.78% with RF. Yang and Guan[10] proposed a feature-optimization approach combined with a SMOTE-XGBoost algorithm to address class imbalance in heart disease prediction. Saboor et al.[11] experimented with nine different machine learning classifiers with hyperparameter tuning and dataset normalization, finding that SVM achieved the best accuracy of 96.72%. Kataria and colleagues[12] found Logistic Regression to be the most accurate model, reaching 93.40% accuracy and outperforming other tested algorithms. Shah et al.[13] obtained the lowest error rates and the highest accuracy (90.79%) on the Cleveland dataset using KNN. Li et al.[14] proposed an EPSOSA-BP neural network integrating particle swarm optimization, simulated annealing, and a feature-elimination mechanism, achieving 93.22% accuracy on the UCI dataset and 95.20% on the Kaggle dataset. On the ensemble side, Mohapatra et al.[15] used a two-level stacking technique with heterogeneous learners, resulting in 92% accuracy; Kavitha et al.[16] used regression and classification techniques in the preprocessing stage on the Cleveland dataset and achieved an overall accuracy of 88.7%. Beyond heart disease specifically, explainable ensemble approaches have also proven effective in adjacent clinical prediction tasks; for instance, Alpsalaz et al.[17] proposed a reliability-weighted fuzzy ensemble with integrated explainability for cervical cancer classification, reflecting a growing consensus that ensemble learning and interpretability are complementary rather than competing priorities in clinical ML.

It is important to note that two methodological gaps can be found in much of this literature. First, class imbalance (when the number of negative (no-disease) samples far exceeds the number of positive (disease) samples) is not always considered; some studies use only a single balancing method with the assumption that class boundaries are always clean and distinct, an assumption evaluated directly by,[18] who compared the statistical performance of three different sampling approaches for handling binary imbalanced datasets. Second, and more importantly, many reported accuracy numbers in this space are hard to confirm with confidence, as the validation procedures are not always clearly described, and there may be leakage of information in the way that the data was split, feature scaling, resampling, or hyperparameter optimization was done. This is a well-known pitfall of tabular clinical ML in general[19–21] and can result in impressive accuracy numbers that do not necessarily reflect a model's true generalization performance to new patients.

This study directly tackles both these gaps. We integrate and pool three public heart disease datasets (Kaggle, UCI, Cleveland), and propose a hybrid class-balancing strategy that incorporates Instance Hardness Threshold (IHT) and Tomek Links, which tackles both class imbalance and noisy class boundaries. Most importantly, we adopt and explicitly state a fold-wise validation protocol that ensures feature scaling, hyperparameter tuning (using nested cross validation), and balancing are carried out within each training fold and no preprocessing step is ever applied to held out test data, which is hardly always the case as summarized in the previous literature. Evaluation is extended beyond raw accuracy to include model calibration, clinical decision curve analysis and a sensitivity–specificity threshold analysis, as we realized that accuracy of classification was an incomplete measure of clinical utility. Lastly, we use SHAP to interpret feature contributions to the result and also pair this with a feature ablation study. Two limitations of the merged dataset arose during this work that we also disclose, but do not obscure, in the Limitations section as suggestions for future work: record-level overlap in the three datasets and the lack of external validation.

The study's primary contributions are listed below:

Developed a hybrid machine learning model for CVD prediction based on XAI, using interpretable methods and hybrid data balancing.
Combined three publicly available datasets into a single pooled dataset to train and evaluate the model.

Applied an IHT + Tomek Links hybrid approach for data balancing, performed strictly within a leakage-controlled, fold-wise validation pipeline.
Trained and compared twelve machine learning classifiers, with hyperparameter tuning nested within cross-validation to find the best attribute values for each classifier, and all metrics reported as mean ± standard deviation and 95% confidence intervals across folds.
Combined the best-performing classifiers using soft- and hard-voting ensemble approaches to obtain the optimal result.
Extended evaluation beyond accuracy to include model calibration (Brier score), clinical decision-curve analysis, and sensitivity–specificity threshold analysis.
Applied SHAP for model explainability, complemented by a feature ablation study quantifying each predictor's individual contribution.

2. Materials and Methods

Figure 1 summarizes the overall workflow adopted in this study. The three source datasets are first merged into a single pooled dataset, which is then partitioned via stratified 5-fold cross-validation *before* any preprocessing is performed. Within each fold, feature scaling, class balancing, and hyperparameter tuning are applied strictly to the training partition; the held-out test partition remains untouched until final evaluation. The best-performing classifiers are then combined into a voting ensemble, and the resulting model is evaluated using both standard discrimination metrics and additional clinical-utility analyses (calibration, decision-curve analysis, and threshold analysis), followed by SHAP-based interpretability and a feature ablation study. Each stage of this pipeline is described in detail in the following subsections.

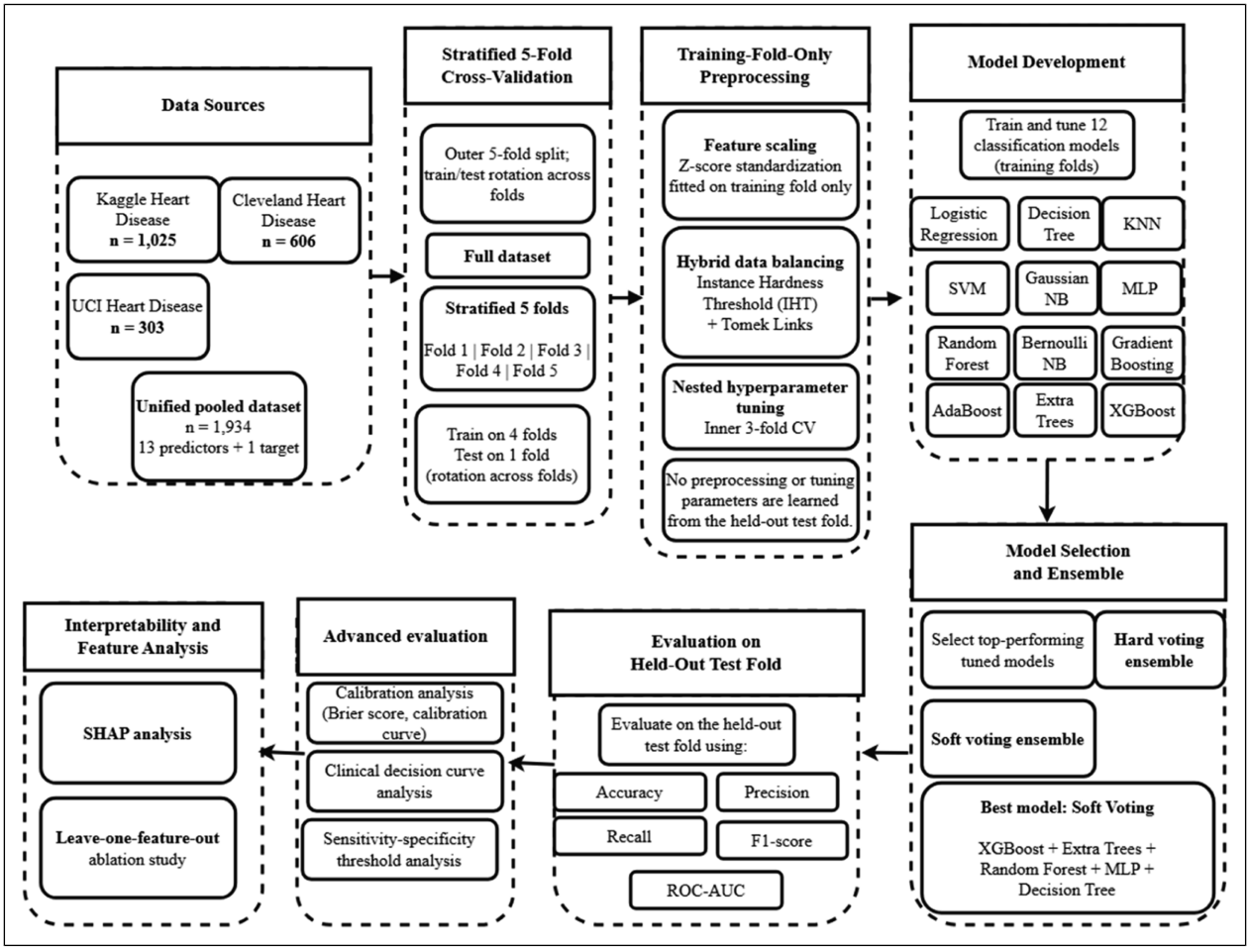


Figure 1. Overview of the proposed methodology, showing the corrected fold-wise validation protocol in which data splitting precedes all preprocessing, balancing, and hyperparameter tuning, restricted to each fold's training partition.

2.1. Data Sets

Three publicly available datasets were used to develop and evaluate the proposed heart disease prediction framework: the Heart Disease Dataset from Kaggle,[22] the UCI Heart Disease Dataset,[23] and the Heart Disease Cleveland Dataset.[24] Each source dataset shares an identical 14-attribute schema, described in Table 1. The three sources were concatenated into a single combined dataset comprising 1,934 records, with 13 predictor features and one binary target variable, used directly for all subsequent modeling. Table 2 summarizes the number of attributes and total entries contributed by each source dataset.

It should be noted that publicly available cardiovascular datasets of this kind are frequently drawn from overlapping or related clinical repositories. Consistent with this, a non-trivial degree of record-level overlap was identified across the three source datasets used in this study. This overlap is discussed further in the Limitations section, and reported performance metrics should be interpreted with this consideration in mind. Feature alignment across the three sources was straightforward given that all originate from the same underlying schema, requiring only matching of column names and confirmation of consistent categorical encodings (e.g., for *cp*, *restecg*, *slope*, and *thal*) prior to concatenation; no missing values were identified across any of the 13 predictor features in the merged dataset. A sensitivity analysis evaluating performance separately on each of the three source datasets, rather than only on the combined pool, was not performed in this study and is identified as a direction for future work in the Limitations section.

2.2. Data Transformation and Scaling

All predictor features were standardized using a z-score transformation so that features measured on different scales (e.g., cholesterol in mg/dL versus categorical slope values) contributed proportionately to distance-based and gradient-based learning algorithms.[25] Each feature x was transformed according to Equation (1):

$$Z = \frac{x - \mu}{\sigma} \tag{1}$$

where μ and σ denote the mean and standard deviation of the feature, respectively. To prevent information leakage between training and test partitions, the scaling parameters (μ, σ) were estimated exclusively from the training partition of each cross-validation fold, and the corresponding held-out test partition was scaled using the same parameters without re-estimation. Figure 2 illustrates the distributions of the predictor variables after z-score standardization, while box plots comparing feature distributions before and after standardization are presented in Figures 3 and 4, respectively.

2.3. Data Splitting and Validation Strategy

For preventing information leakage from preprocessing, resampling and hyperparameter selection before data splitting, the proposed processing pipeline is a fold-wise processing pipeline, which was built around the StratifiedKFold, shuffled, fixed random seed for data reproducibility, as used in stratified 5-fold cross validation. The data sets were initially divided into five outer folds and then each outer fold was further split into training and testing sets, maintaining the class ratio in each set as it

Table 1. Detailed overview of clinical attributes, operational encodings, and data types for the 14 consolidated features within the unified heart disease prediction framework.

#	Feature	Description	Type
1	age	Age of the patient in years	int64
2	sex	Biological sex (1 = male, 0 = female)	int64
3	cp	Chest pain type (0=typical, 1=atypical, 2=non-anginal, 3=asymptomatic)	int64
4	trestbps	Resting blood pressure on admission (mm Hg)	int64
5	chol	Serum cholesterol level (mg/dL)	int64
6	fbs	Fasting blood sugar > 120 mg/dL (1 = true, 0 = false)	int64
7	restecg	Resting ECG results (0=normal, 1=ST-T wave, 2=probable/definite LVH)	int64
8	thalach	Maximum heart rate achieved during exercise	int64
9	exang	Exercise-induced angina (1 = yes, 0 = no)	int64
10	oldpeak	ST depression induced by exercise relative to rest	float64
11	slope	Slope of the peak exercise ST segment (0=up, 1=flat, 2=down)	int64
12	ca	Number of major vessels (0–3) colored by fluoroscopy	int64
13	thal	Thalassemia status (1 = normal, 2 = fixed defect, 3 = reversible defect)	int64
14	target	Presence of heart disease (1 = disease, 0 = no disease)	int64

Table 2. Dataset Summary.

Dataset Source	Attributes	Total Entries
Kaggle Heart Disease[22]	14	1025
UCI Heart Disease[23]	14	606
Cleveland Heart Disease[24]	14	303
Combined Pool	**14**	**1934**

existed in the original data set. All further processing (feature scaling, class balancing, hyperparameter optimization) was performed on the training portion of that fold only, with the test portion of the fold never seen by any fitting procedure and solely used for final prediction and evaluation. The hyperparameter optimization was done using a grid search within each outer fold of training, while an inner 3-fold cross validation loop was used, meaning that the model selection was never based on data from the outer fold used for evaluation. The five outer folds are used to report final performance metrics as mean, standard deviation, and 95% confidence interval.[26]

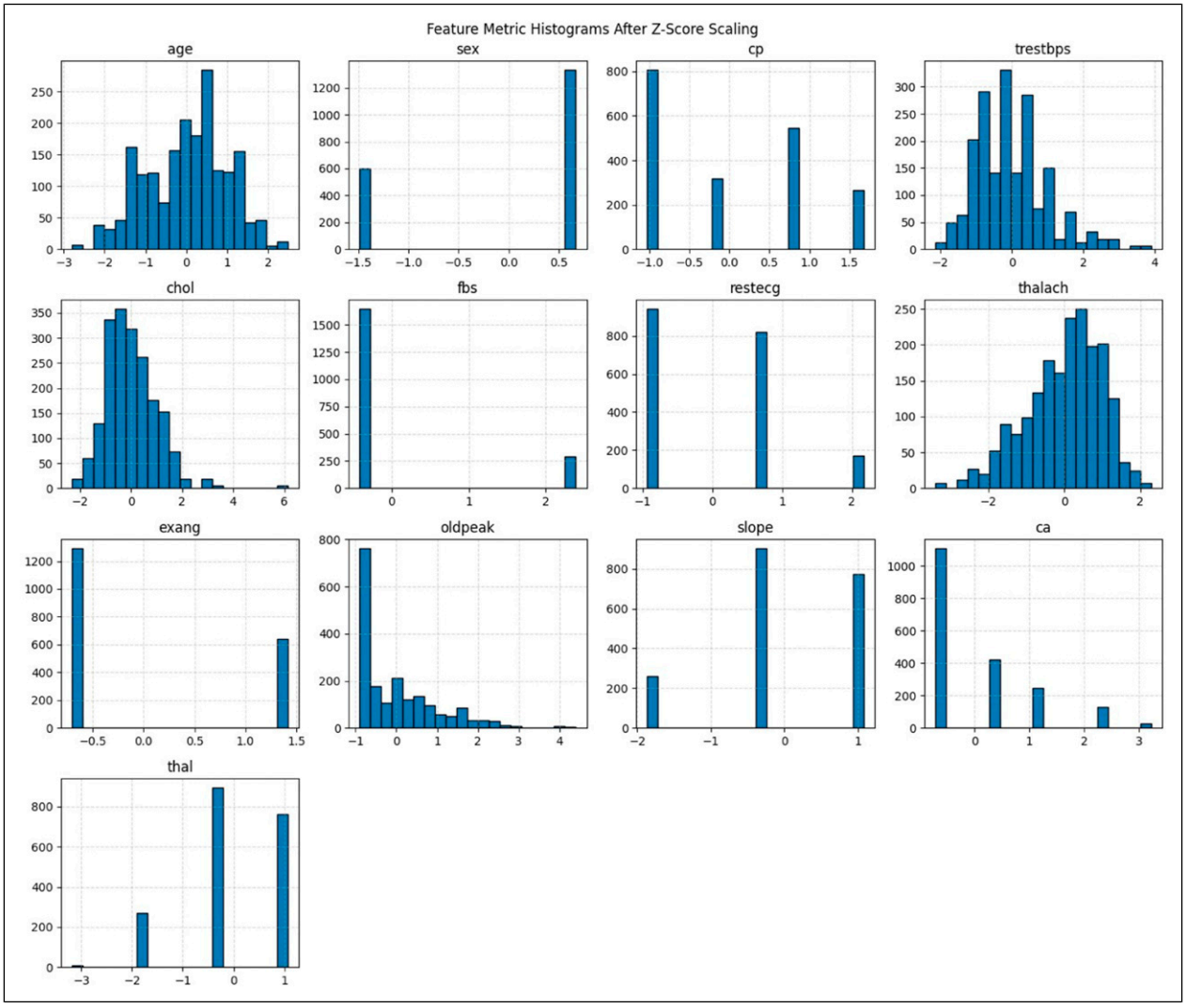


Figure 2. Histograms illustrating the standardized feature distributions across the consolidated dataset after z-score scaling, showing the normalized ranges and distribution patterns of the predictor variables.

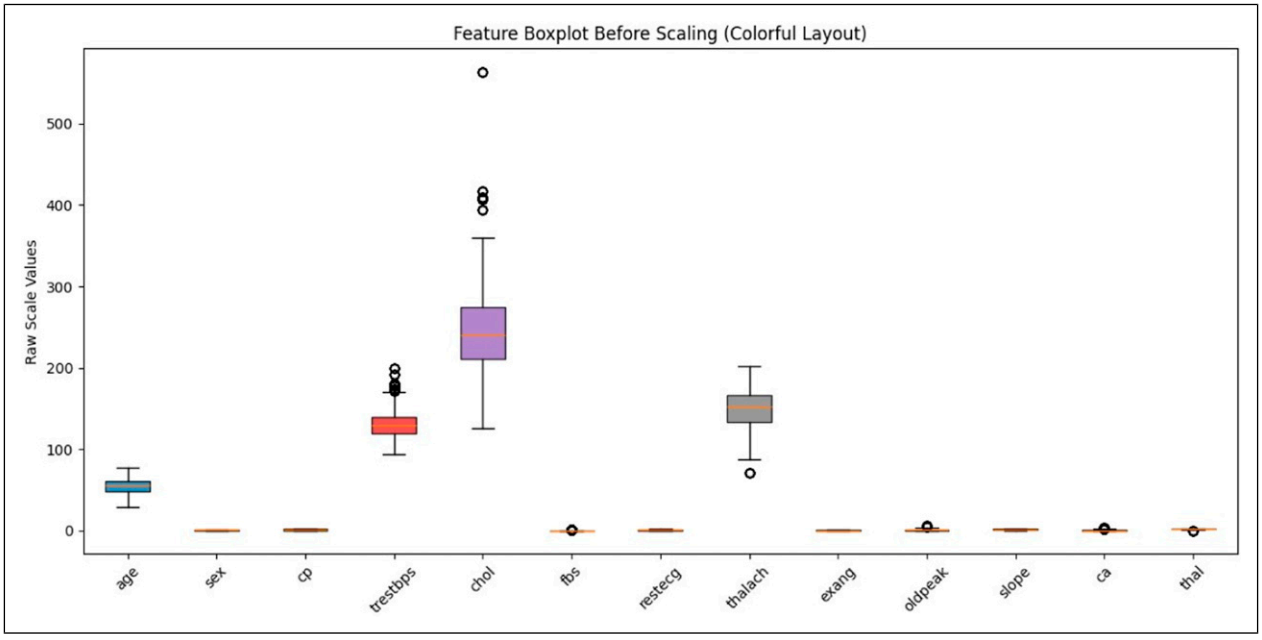


Figure 3. Box plots of predictor features before z-score standardization, highlighting the significant disparities in feature scales and variances.

2.4. Data Balancing

Two complementary techniques were used to address class imbalance and class-boundary noise. The Instance Hardness Threshold (IHT)[18] technique estimates the difficulty of correctly classifying each majority-class instance using the prediction error of a base classifier, and removes instances whose predicted difficulty exceeds a defined threshold. For each sample x_i, the instance hardness is computed as shown in Equation (2):

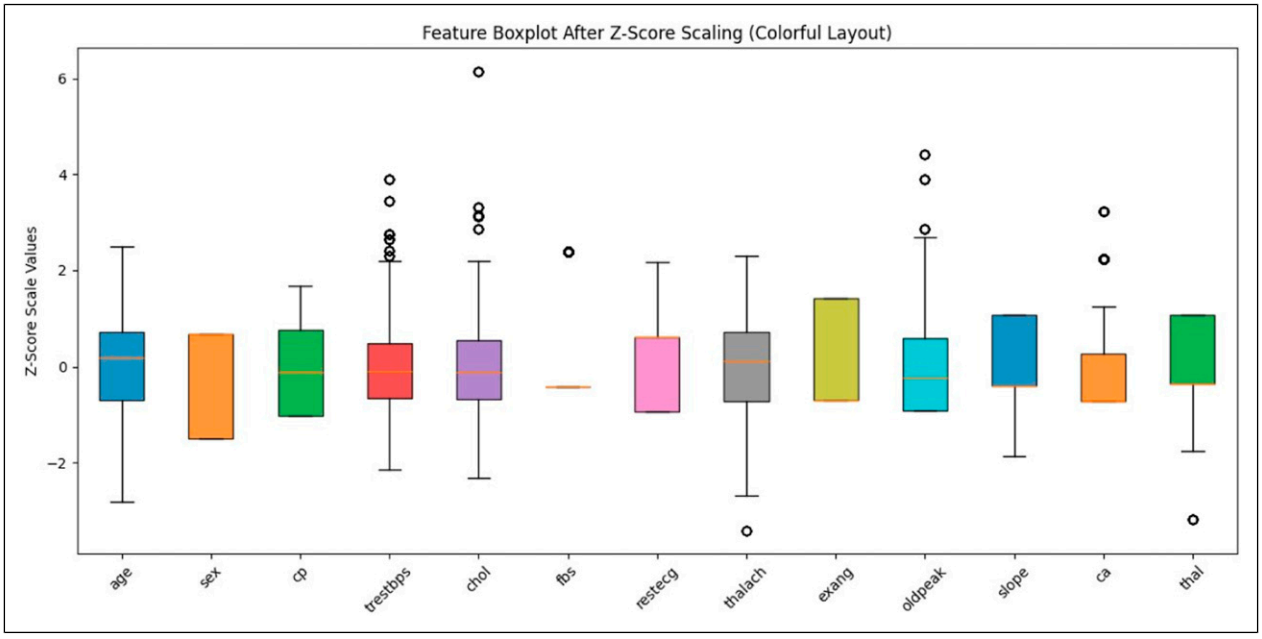


Figure 4. Box plots of predictor features after z-score standardization, demonstrating the uniform scaling centering all features around a mean of zero with unit variance.

$$IH(x_i) = 1 - P(y_i \mid x_i) \tag{2}$$

where $P(y_i \mid x_i)$ denotes the predicted probability of the true class label. Majority-class samples with $IH(x_i)$ greater than threshold θ were removed.

Tomek Links[18] were then applied to the IHT-filtered training data to remove residual noisy boundary points. A Tomek Link exists between samples x_i and x_j of different classes when each is the other's nearest neighbor, defined mathematically by Equation (3):

$$\mathrm{NN}(x_i) = x_j, \quad \mathrm{NN}(x_j) = x_i, y_i \neq y_j \tag{3}$$

For each identified pair, the majority-class member was removed. The resulting balanced training set is expressed as detailed in Equation (4):

$$D_{\text{balanced}} = \mathrm{Tomek}(\{x_i \in D \mid IH(x_i) \leq \theta\}) \tag{4}$$

For the merged dataset, both global class imbalance and global class overlap were present, thus a hybrid combination of IHT and Tomek Links was used. As in the validation of the fold-wise strategy, this balancing process was carried out only in the training partition of each cross validation fold, while the test partition of each fold remained in its original, imbalanced, clinical state, allowing the reported performance to reflect generalization to realistic, imbalanced clinical data.

2.5. Machine Learning Classifiers

Twelve machine learning algorithms from six computational paradigms were implemented and tested to isolate the best predictive architecture. Linear and instance-based baselines were ensembled as L_2-regularized Logistic Regression to model linear log-odds, Support Vector Machines (SVM) with a Radial Basis Function (RBF) kernel to learn non-linear boundaries by hyper-planar margin maximization, and K-Nearest Neighbors (KNN) to compute local majority votes based on an optimized Minkowski distance metric. Two probabilistic models, Gaussian Naive Bayes and Bernoulli Naive Bayes, were baseline models using maximum likelihood estimation with conditional independence for continuous and binary features, respectively.

A standard non-parametric Decision Tree (DT) approach using recursive Gini impurity splits was used to baseline the structural decision logic. In order to reduce variance and overfitting, parallel bagging ensembles were implemented based on Random Forest and Extra Trees (Extremely Randomized Trees), the latter using completely randomized threshold selections. Sequential boosting frameworks were extensively used to iteratively minimize the exponentially loss function by reweighting instances (AdaBoost); to optimize a differentiable version of the cross-entropy loss by following gradient vectors (standard Gradient Boosting); and to optimize a regularized objective function using second-order Taylor expansions (XGBoost). Finally, a Multi-Layer Perceptron (MLP) was realized as a connectionist neural network, using the Adam algorithm to optimize the weights of the hidden layers, and treating the problem of error backpropagation.[27]

2.6. Hyperparameter Tuning

Each classifier's hyperparameters were optimized via grid search (GridSearchCV), nested within the training partition of each outer cross-validation fold, using an inner 3-fold cross-validation loop to select the best-performing configuration. This nested procedure was repeated independently within each of the five outer folds, so that selected hyperparameters, and the resulting performance estimate, reflect only information available within that fold's training data.[28] The full search grids evaluated for each classifier are reported in Table 3, and representative best-parameter configurations identified across folds are summarized in Table 4.

2.7. Ensemble Approach

After individual model evaluation, the best-performing base classifiers (by mean cross-validated accuracy) were aggregated in all possible two or more model combinations using either soft voting (in which the probabilities for the various classes predicted by the constituent models were averaged and a threshold set at 0.5) or hard voting (where the final class prediction was made by majority vote of the constituent models' predictions). Each ensemble combination was tested on the same fold-wise cross validation protocol used for each classifier, and the mean accuracy with 95% confidence intervals were calculated.[29]

Table 3. Hyperparameter Search Grids.

Model	Parameter Grid Searched
Logistic Regression	C: [0.01, 0.1, 1, 10]; solver: ['lbfgs', 'liblinear']
Decision Tree	max_depth: [5, 10, 15, 20, None]; min_samples_split: [2, 5, 10]
SVM	C: [0.1, 1, 10]; kernel: ['rbf', 'linear']; gamma: ['scale', 'auto']
KNN	n_neighbors: [3, 5, 7, 9, 11]; weights: ['uniform', 'distance']
Gaussian Naive Bayes	No tunable hyperparameters searched (default var_smoothing used)
Bernoulli Naive Bayes	alpha: [0.1, 0.5, 1.0, 2.0]
Random Forest	n_estimators: [100, 200, 300]; max_depth: [10, 20, None]
Gradient Boosting	learning_rate: [0.05, 0.1, 0.2]; n_estimators: [100, 200, 300]
AdaBoost	learning_rate: [0.5, 1.0]; n_estimators: [100, 200]
Extra Trees	n_estimators: [100, 200, 300]; max_depth: [10, 20, None]
XGBoost	learning_rate: [0.05, 0.1, 0.2]; n_estimators: [100, 200]
MLP	activation: ['relu', 'tanh']; hidden_layer_sizes: [(50,), (100,)]; solver: ['adam']

2.8 Model Evaluation Metrics

Model performance was quantified using accuracy, precision, recall, F_1-score, and area under the ROC curve (AUC), which are formally defined in Equations (5) to (8):

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}} \tag{5}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

$$F_1\text{ - score} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \tag{8}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. All metrics are reported as the mean ± standard deviation across the five cross-validation folds, together with 95% confidence intervals.[30]

Table 4. Optimized Hyperparameter Settings for Base Classifiers.

Model	Optimal Hyperparameter Configuration
Logistic Regression	C: 0.1, solver: 'lbfgs'
Decision Tree	max_depth: None, min_samples_split: 2
SVM	C: 10, gamma: 'scale', kernel: 'rbf'
KNN	n_neighbors: 5, weights: 'distance'
Gaussian Naive Bayes	Default parameters used
Bernoulli Naive Bayes	alpha: 0.1
Random Forest	max_depth: None, n_estimators: 200
Gradient Boosting	learning_rate: 0.1, n_estimators: 300
AdaBoost	learning_rate: 1.0, n_estimators: 200
Extra Trees	max_depth: None, n_estimators: 100
XGBoost	learning_rate: 0.1, n_estimators: 100
MLP	activation: 'tanh', hidden_layer_sizes: (100,), solver: 'adam'

Table 5. Software and Platform Environment.

Component	Detail
Platform	Kaggle Notebook, standard CPU runtime
Language	Python 3.11
Data handling	pandas 2.2.2, NumPy 1.26.4
Classical ML, validation	scikit-learn 1.5.2
Gradient boosting	XGBoost 3.3.0
Class balancing	imbalanced-learn 0.14.2
Interpretability	SHAP 0.46.0
Visualization	Matplotlib 3.9.2, seaborn 0.13.2

2.9. Calibration Analysis

Beyond discriminative performance, the clinical reliability of a predictive model depends on how well its predicted probabilities correspond to observed outcome frequencies. To assess this, the Brier score was computed for the top-performing model across folds, defined as shown in Equation (9):

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}\left(p_i - o_i\right)^2 \tag{9}$$

where p_i is the predicted probability of the positive class for sample i, o_i is the observed outcome (1 if the sample is a true positive case, 0 otherwise), and N is the total number of samples. Lower Brier scores indicate better-calibrated probability estimates. A probability calibration curve was additionally constructed comparing predicted probability against observed event frequency across probability bins. Results are presented in the Results section.[31]

2.10. Clinical Decision Curve Analysis

To evaluate the clinical utility of the top-performing model beyond standard discrimination metrics, decision curve analysis (DCA) was performed. Net benefit at a given probability threshold p_t was calculated using Equation (10):

$$\mathrm{NB}\left(p_t\right) = \frac{\mathrm{TP}}{N} - \left(\frac{\mathrm{FP}}{N}\right) \times \left(\frac{p_t}{1-p_t}\right) \tag{10}$$

where TP and FP denote the number of true positives and false positives at threshold p_t, and N is the total sample size. Net benefit was calculated across a range of probability thresholds and compared against "treat all" and "treat none" reference strategies. The sensitivity–specificity trade-off of the top-performing model was additionally examined across the full range of classification thresholds. Results for both analyses are presented in the Results section.[32]

Table 6. Baseline Classifier Performance (5-Fold CV, IHT+Tomek, No Tuning).

Model	Accuracy	Precision	Recall	F_1-Score	Brier Score
XGBoost	95.66% ± 0.64%	97.91% ± 0.87%	93.57% ± 1.36%	95.68% ± 0.66%	0.0386 ± 0.0050
Random Forest	95.55% ± 0.84%	97.61% ± 1.16%	93.67% ± 1.51%	95.59% ± 0.85%	0.0332 ± 0.0046
Extra Trees	95.50% ± 1.11%	97.70% ± 1.26%	93.47% ± 1.56%	95.53% ± 1.12%	0.0298 ± 0.0044
MLP	95.45% ± 0.90%	97.50% ± 0.81%	93.57% ± 1.99%	95.48% ± 0.94%	0.0403 ± 0.0091
Decision Tree	95.14% ± 1.17%	96.68% ± 0.92%	93.77% ± 1.51%	95.20% ± 1.18%	0.0486 ± 0.0117
Gradient Boosting	92.40% ± 0.95%	94.26% ± 0.55%	90.75% ± 1.79%	92.46% ± 1.00%	0.0712 ± 0.0037
KNN	91.16% ± 1.50%	92.30% ± 1.34%	90.35% ± 1.78%	91.31% ± 1.49%	0.0739 ± 0.0106
SVM	90.90% ± 1.12%	92.81% ± 1.41%	89.25% ± 1.81%	90.98% ± 1.14%	0.0720 ± 0.0048
AdaBoost	79.94% ± 1.93%	82.12% ± 2.59%	78.09% ± 2.63%	80.02% ± 1.91%	0.1873 ± 0.0030
Gaussian NB	75.49% ± 2.34%	75.54% ± 3.30%	77.69% ± 1.21%	76.56% ± 1.90%	0.1885 ± 0.0148
Logistic Regression	76.42% ± 2.24%	76.85% ± 2.41%	77.59% ± 2.49%	77.20% ± 2.10%	0.1624 ± 0.0102
Bernoulli NB	72.75% ± 1.14%	73.34% ± 0.72%	73.87% ± 2.01%	73.60% ± 1.35%	0.1904 ± 0.0078

Table 7. Classifier Performance After Nested Hyperparameter Tuning.

Model	Accuracy	Precision	Recall	F_1-Score	Brier Score
XGBoost	95.86% ± 0.85%	97.82% ± 1.00%	94.07% ± 1.66%	95.90% ± 0.86%	0.0369 ± 0.0064
Extra Trees	95.76% ± 1.14%	97.91% ± 1.19%	93.77% ± 1.64%	95.79% ± 1.14%	0.0296 ± 0.0043
Random Forest	95.55% ± 0.84%	97.71% ± 1.25%	93.57% ± 1.40%	95.58% ± 0.84%	0.0330 ± 0.0048
MLP	95.29% ± 0.76%	97.30% ± 1.03%	93.47% ± 1.46%	95.33% ± 0.77%	0.0417 ± 0.0084
Decision Tree	95.14% ± 1.17%	96.68% ± 0.92%	93.77% ± 1.51%	95.20% ± 1.18%	0.0486 ± 0.0117
SVM	95.14% ± 1.05%	96.68% ± 0.70%	93.77% ± 1.70%	95.20% ± 1.08%	0.0411 ± 0.0068
Gradient Boosting	95.09% ± 0.71%	96.88% ± 0.72%	93.47% ± 1.23%	95.14% ± 0.72%	0.0433 ± 0.0049
KNN	93.74% ± 1.47%	94.97% ± 1.78%	92.76% ± 1.41%	93.85% ± 1.44%	0.0480 ± 0.0111
AdaBoost	81.18% ± 1.38%	83.28% ± 1.72%	79.40% ± 2.25%	81.27% ± 1.44%	0.2064 ± 0.0034
Gaussian NB	75.49% ± 2.34%	75.54% ± 3.30%	77.69% ± 1.21%	76.56% ± 1.90%	0.1885 ± 0.0148
Logistic Regression	74.46% ± 1.87%	74.70% ± 2.45%	76.28% ± 1.16%	75.46% ± 1.52%	0.1836 ± 0.0127
Bernoulli NB	72.75% ± 1.14%	73.34% ± 0.72%	73.87% ± 2.01%	73.60% ± 1.35%	0.1904 ± 0.0078

2.11. Feature Ablation Study

To assess the individual contribution of each predictor feature to model performance, a leave-one-feature-out ablation analysis was conducted for the top-performing model. For each feature j, the model was retrained and re-evaluated on the feature set with j removed, and the resulting performance change was computed as stated in Equation (11):

$$\Delta \mathrm{Acc}_j = \mathrm{Acc}_{\mathrm{full}} - \mathrm{Acc}_{-j} \tag{11}$$

where $\mathrm{Acc}_{\mathrm{full}}$ is the mean cross-validated accuracy using the complete feature set, and Acc_{-j} is the mean cross-validated accuracy with feature j excluded, evaluated using the same fold-wise cross-validation protocol described above. A positive $\Delta \mathrm{Acc}_j$ indicates that feature j contributes positively to model performance; a negative value indicates its removal did not harm, or slightly improved, performance. The equivalent computation was applied to F_1-score. Results are presented in the Results section.

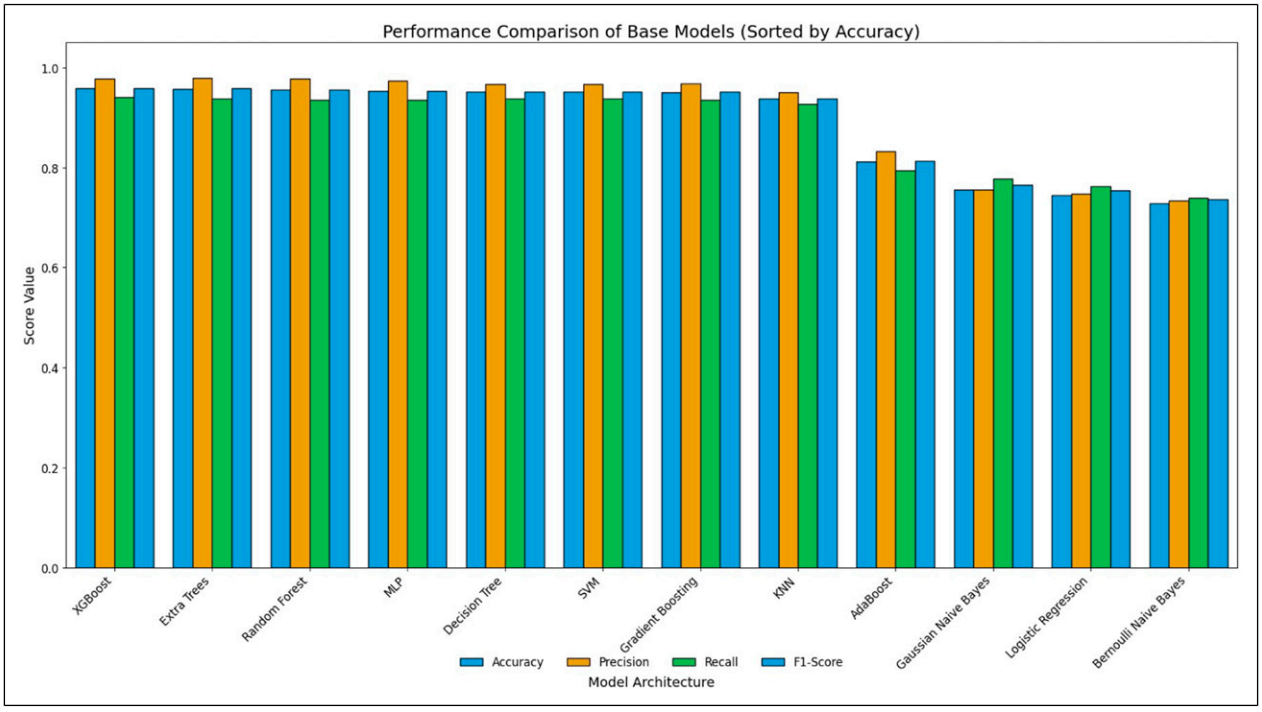


Figure 5. Performance comparison of the evaluated machine learning classifiers using accuracy, precision, recall, and F1-score, with models ordered by predictive performance.

Table 8. Top 5 Soft Voting and Top 5 Hard Voting Combinations.

Rank	Voting Combination	Accuracy (95% CI)
Soft Voting Configurations		
1	XGB+ET+RF+MLP+DT	96.02% ± 0.62% [95.47%, 96.56%]
2	XGB+ET+MLP+DT	95.97% ± 0.70% [95.35%, 96.58%]
3	XGB+ET	95.92% ± 0.85% [95.17%, 96.66%]
4	XGB+ET+RF	95.92% ± 0.72% [95.29%, 96.54%]
5	XGB+RF+MLP+DT	95.92% ± 0.62% [95.37%, 96.46%]
Hard Voting Configurations		
1	XGB+ET+DT	95.92% ± 1.04% [95.01%, 96.82%]
2	XGB+RF+DT	95.92% ± 0.94% [95.09%, 96.74%]
3	XGB+MLP+DT	95.81% ± 0.72% [95.18%, 96.44%]
4	ET+RF+DT	95.76% ± 1.16% [94.74%, 96.78%]
5	ET+MLP+DT	95.76% ± 1.08% [94.82%, 96.71%]

2.12. Interpretability

SHAP (SHapley Additive exPlanations)[33–35] was applied to the top-performing constituent model to quantify the contribution of each feature to individual predictions, based on the Shapley value from cooperative game theory. For a model f and feature set F, the SHAP value of feature i for a given prediction is defined by Equation (12):

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \cdot [f(S \cup \{i\}) - f(S)] \tag{12}$$

where S denotes a subset of features excluding feature i, and $f(S)$ denotes the model's expected prediction using only the features in S. This value represents feature i's average marginal contribution to the prediction across all possible feature subsets, providing a theoretically grounded decomposition of each prediction into additive per-feature contributions. This analysis supports transparency and clinical interpretability of the proposed framework; corresponding results are presented in the Results section.

2.13. Experimental Setup

All experiments were implemented in Python and executed on Kaggle Notebook, using the standard Kaggle CPU runtime. Table 5 summarizes the software environment. The study, including dataset acquisition, data harmonization, preprocessing, model development, cross-validation, ensemble evaluation, explainability analysis, and preparation of the final experimental results, was conducted from 01 July 2025 to 28 December 2025.

Table 9. Top 10 Ensemble Configurations, Full Metrics.

Rank	Model	Accuracy	Precision	Recall	F_1-Score
1	Soft (XGB+ET+RF+MLP+DT)	96.02%	97.83%	94.37%	96.06%
2	Soft (XGB+ET+MLP+DT)	95.97%	97.83%	94.27%	96.00%
3	Hard (XGB+ET+DT)	95.92%	98.02%	93.97%	95.94%
4	Hard (XGB+RF+DT)	95.92%	98.12%	93.87%	95.94%
5	Soft (XGB+ET+RF)	95.92%	97.73%	94.27%	95.95%
6	Soft (XGB+ET)	95.92%	97.82%	94.17%	95.95%
7	Soft (XGB+RF+MLP+DT)	95.92%	97.83%	94.17%	95.95%
8	Hard (XGB+MLP+DT)	95.81%	97.72%	94.07%	95.85%
9	Hard (ET+RF+DT)	95.76%	97.92%	93.77%	95.79%
10	Hard (ET+MLP+DT)	95.76%	98.02%	93.67%	95.78%

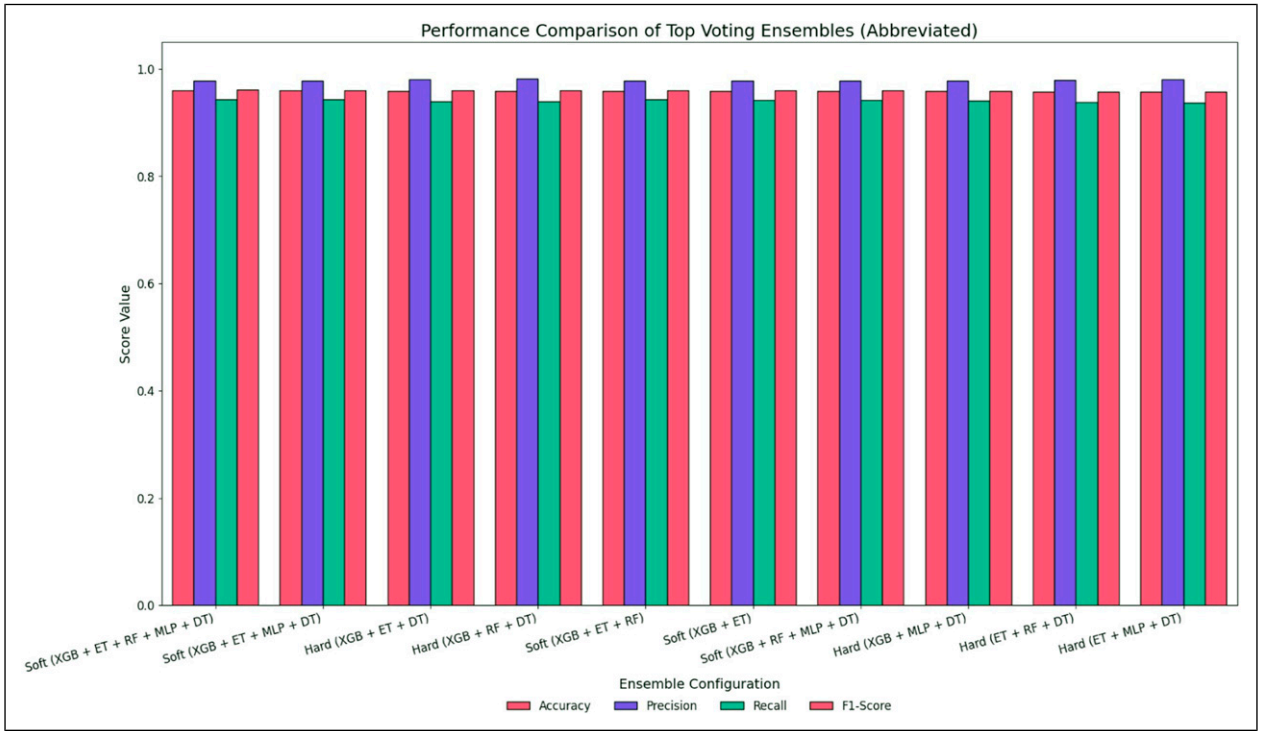


Figure 6. Graphical comparison of predictive performance metrics across the top-performing soft-voting and hard-voting ensemble configurations.

3. Results

Model performance was evaluated using stratified 5-fold cross-validation with fold-wise preprocessing, class balancing, and nested hyperparameter tuning, as described in the Methods section. All metrics below are reported as the mean across folds, with standard deviation and 95% confidence intervals where applicable.

3.1. Baseline Classifier Performance (Before Hyperparameter Tuning)

Table 6 presents the performance of the twelve base classifiers under fold-wise IHT + Tomek Links balancing, prior to hyperparameter optimization. XGBoost achieved the highest baseline accuracy (95.66%), closely followed by Random Forest (95.55%), Extra Trees (95.50%), MLP (95.45%), and Decision Tree (95.14%). The three probabilistic/linear models — Logistic Regression, Gaussian Naive Bayes, and Bernoulli Naive Bayes — formed a distinct lower-performing cluster (72–76% accuracy), consistent with their limited capacity to capture non-linear feature interactions in this dataset. AdaBoost also underperformed relative to the tree-ensemble family (79.94%).

3.2. Classifier Performance After Hyperparameter Tuning

Table 7 shows performance after nested hyperparameter optimization. Gains over the untuned baseline were modest for most classifiers (generally under 0.5 percentage points), with XGBoost improving from 95.66% to 95.86% and Extra Trees from 95.50% to 95.76%. Logistic Regression showed a small decline (76.42% → 74.46%), likely reflecting fold-to-fold variability at its lower performance ceiling rather than tuning quality. Bernoulli Naive Bayes showed no change from its untuned baseline, indicating the model's performance on this dataset was insensitive to the smoothing parameter α despite grid search selecting a different optimal value ($\alpha = 0.1$) than the default. The relative ranking of classifiers was largely preserved before and after tuning, as visually summarized across the models in Figure 5. The five top-ranked classifiers by mean accuracy — XGBoost, Extra Trees, Random Forest, MLP, and Decision Tree — were carried forward into the ensemble stage.

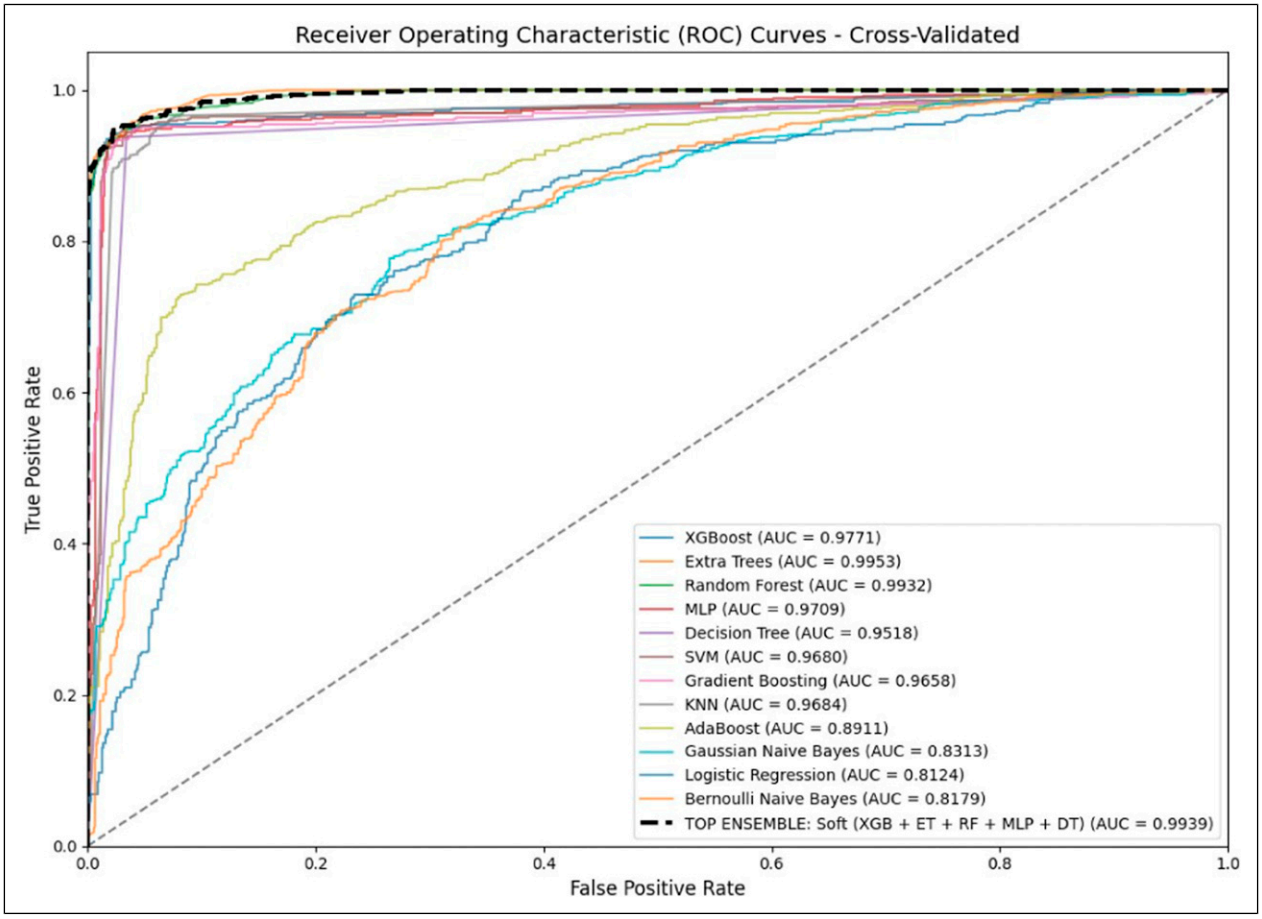


Figure 7. Receiver Operating Characteristic (ROC) curves evaluated across the twelve base classifiers and the top-performing soft-voting ensemble framework, showcasing the true positive rate against the false positive rate across all decision thresholds.

3.3. Voting Ensemble Performance

Combinations of the top five tuned classifiers were evaluated under soft-voting and hard-voting aggregation. Soft voting consistently outperformed hard voting among the top configurations, with the five-model soft-voting ensemble (XGBoost + Extra Trees + Random Forest + MLP + Decision Tree) achieving the highest accuracy of any configuration tested. Performance differences among the top ten combinations were small (95.76%–96.02%), indicating the ensemble result is not driven by a single dominant classifier pairing but is a stable improvement across several related combinations. Table 8 breaks down the top configurations by accuracy performance and corresponding confidence margins, while Table 9 documents full verification metrics for the top 10 models. Figure 6 presents this comparison graphically.

3.4. Overall Top-Performing Model

The best individual classifier (XGBoost, 95.86%) was outperformed by the best ensemble configuration. The soft-voting ensemble of XGBoost, Extra Trees, Random Forest, MLP, and Decision Tree was identified as the overall top-performing model at 96.02% accuracy, and was used for all subsequent model-specific analyses reported below.

3.5. ROC Curve Analysis

Figure 7 presents cross-validated ROC curves for all twelve base classifiers alongside the top-performing ensemble. Extra Trees achieved the highest individual AUC (0.9953), closely followed by Random Forest (0.9932) and the top ensemble itself (0.9939); XGBoost, despite having the highest tuned accuracy, achieved a comparatively lower AUC of 0.9771. MLP, SVM, KNN, and Gradient Boosting formed a mid-range cluster (AUC 0.9658–0.9709), while Decision Tree (0.9518) and

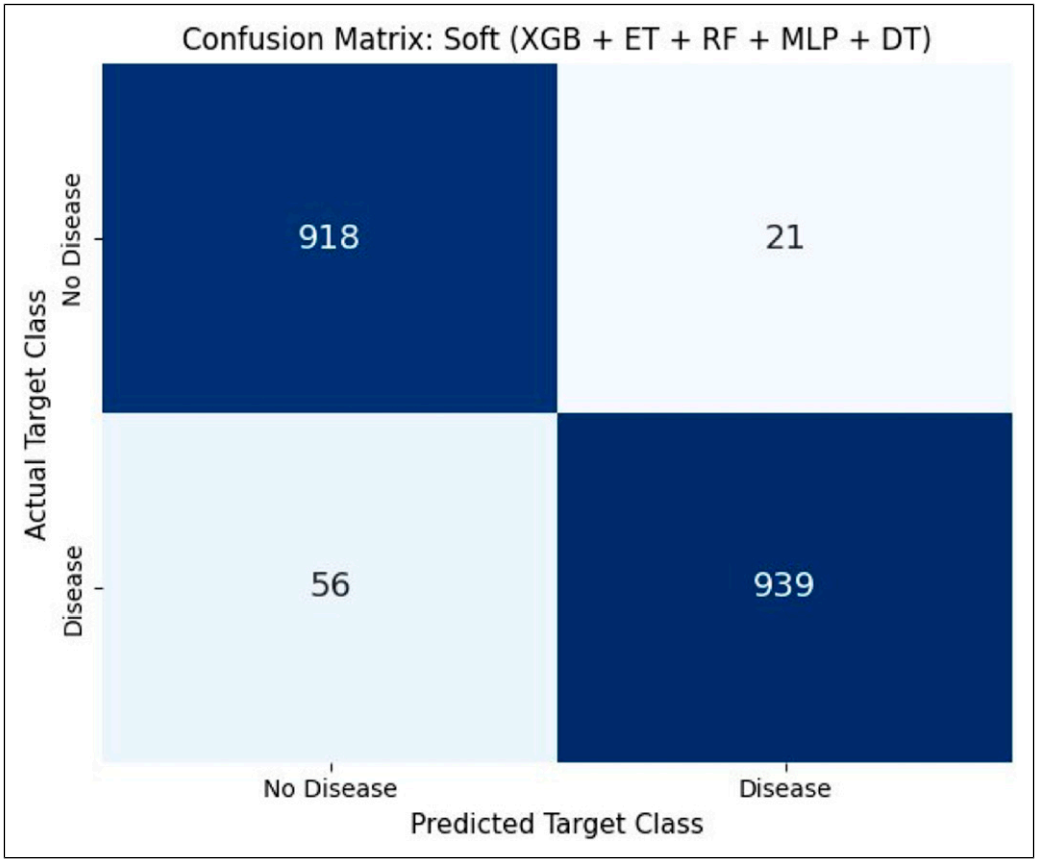


Figure 8. Confusion matrix of the top-performing ensemble framework displaying true positive, true negative, false positive, and false negative counts on out-of-fold predictions.

AdaBoost (0.8911) trailed further behind. The three probabilistic/linear models again performed weakest by this metric (Gaussian NB 0.8313, Bernoulli NB 0.8179, Logistic Regression 0.8124), consistent with their accuracy-based ranking in Tables 6 and 7. The top ensemble's ROC curve sits at or above nearly all individual base-classifier curves across the full false-positive-rate range, indicating consistently strong discrimination rather than an advantage concentrated at one operating point.

3.6. Confusion Matrix of the Top Model

Figure 8 presents the confusion matrix of the top-performing ensemble model on pooled out-of-fold predictions: 918 true negatives, 21 false positives, 56 false negatives, and 939 true positives ($N = 1,934$). These counts reproduce the 96.02% accuracy, 97.83% precision, and 94.37% recall reported in Table 9, and show that the model's errors are not concentrated disproportionately in either direction, false negatives (56) modestly outnumber false positives (21), consistent with the somewhat lower recall relative to precision.

3.7. SHAP Feature Importance

Because the top-performing model is a multi-classifier ensemble, SHAP analysis was applied to its primary constituent, XGBoost, using TreeExplainer. Figure 9 presents the resulting SHAP summary plot. Chest pain type (*cp*) shows the widest spread of SHAP values of any feature, with high values (more severe or asymptomatic chest pain codings) associated with strongly positive SHAP values that push predictions toward the disease class, and low values pulling predictions in the opposite direction. Thalassemia status (*thal*) and ST-segment slope (*slope*) show similarly wide, clearly bidirectional spreads, with abnormal *thal* status and steeper *slope* values generally increasing predicted risk. Number of major vessels (*ca*) and *sex* show moderate, clearly separated spreads, while resting ECG results (*restecg*), ST depression (*oldpeak*), and

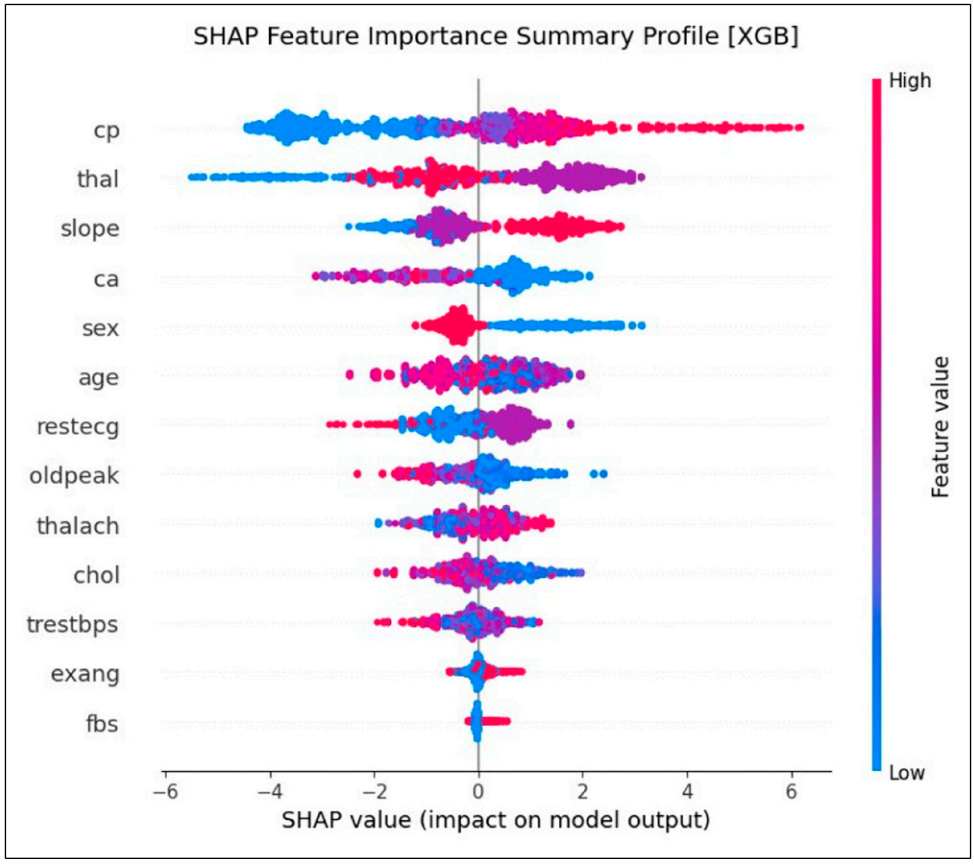


Figure 9. SHAP summary dot plot demonstrating the distribution of impact that individual clinical features assert on the underlying XGBoost model's predictive output.

maximum heart rate (*thalach*) contribute more modestly. Fasting blood sugar (*fbs*) and exercise-induced angina (*exang*) show the narrowest spread of SHAP values, indicating minimal influence on individual predictions for this model.

3.8. Probability Calibration

A calibration curve was constructed comparing the top model's predicted probabilities against observed outcome frequencies across ten probability bins, together with the Brier score. Figure 10 presents this curve. The top ensemble achieved a Brier score of 0.0315, indicating well-calibrated probability estimates overall. The calibration curve tracks the diagonal (perfectly calibrated) line closely at the low (0.0–0.35) and high (0.75–1.0) ends of the predicted-probability range. Some deviation is visible in the mid-range, particularly around predicted probabilities of 0.45–0.65, where the curve dips below and then jumps above the diagonal; this is most likely attributable to smaller sample counts within these middle probability bins rather than a systematic calibration failure.

3.9. Clinical Decision Curve Analysis

Decision curve analysis (DCA) was performed for the top-performing model across probability thresholds from 0.01 to 0.99, compared against "treat-all" and "treat-none" reference strategies. Figure 11 presents the resulting net-benefit curves. The top model's net benefit exceeds the "treat-all" strategy across the entire threshold range tested, with the gap widening substantially as the threshold increases: near threshold 0.4 the "treat-all" strategy's net benefit falls below the "treat-none" baseline (net benefit ≈0), while the model continues to deliver a net benefit of roughly 0.46–0.48 in this same range. From that point onward, the model's curve remains well above both reference strategies for the remainder of the threshold range, declining only gradually and dropping sharply just above threshold 0.95 as very few patients are classified as positive at such stringent cutoffs. This indicates the model provides positive clinical utility relative to both a blanket "treat everyone"

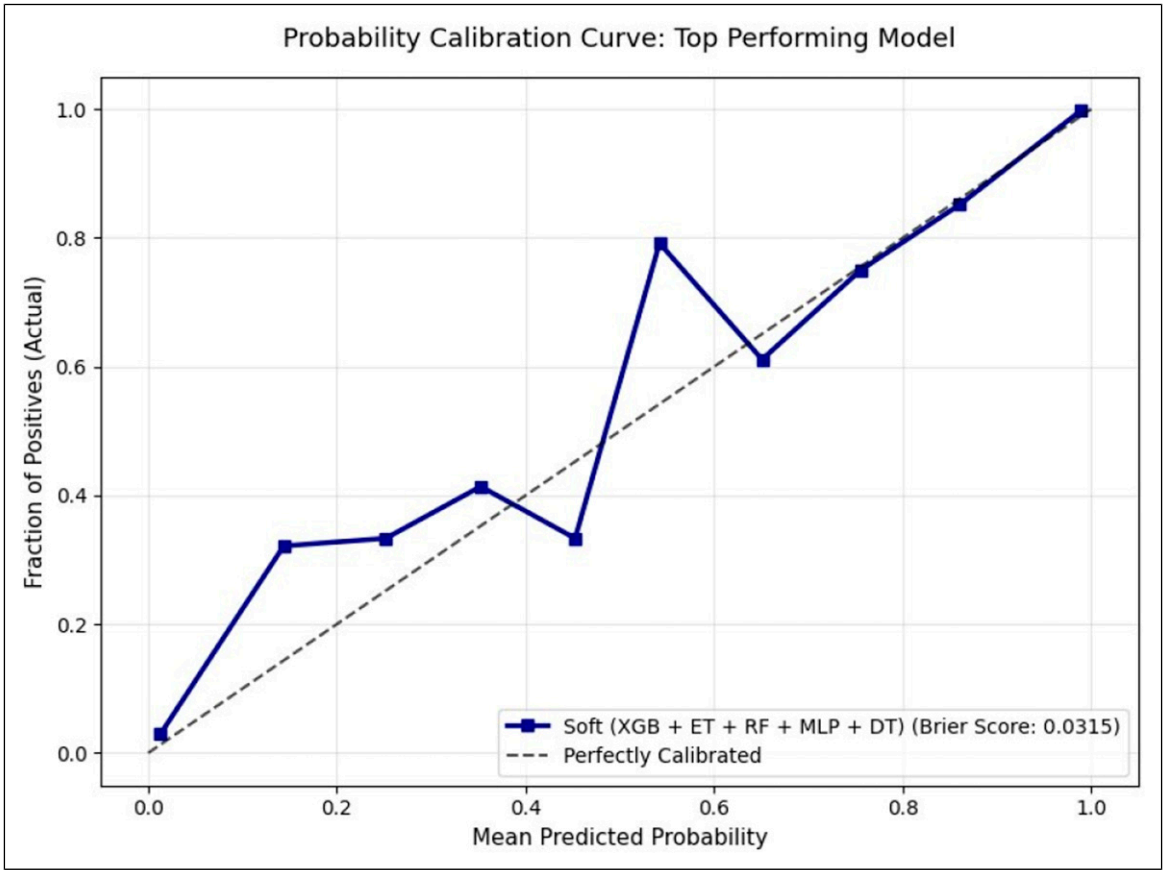


Figure 10. Probability calibration curve with calculated Brier score evaluating the reliability of the ensemble model's predicted risk probabilities against actual observed diagnostic distribution frequencies.

approach and a "treat no one" approach across essentially the full range of clinically plausible decision thresholds, with the largest relative advantage over "treat-all" emerging at moderate-to-high thresholds (roughly 0.4–0.9).

3.10. Sensitivity–Specificity Trade-off

Sensitivity and specificity of the top-performing model were evaluated across the full range of classification thresholds. Figure 12 presents the full trade-off curve, showing sensitivity declining and specificity rising as the decision threshold increases, with the two curves crossing near a decision threshold of 0.32. The optimal balance point, where sensitivity and specificity are most closely matched, occurred at a decision threshold of 0.323, with a sensitivity of 95.68% and a specificity of 95.74%. The near-equal sensitivity and specificity at this threshold indicate the model does not systematically favor detecting positive cases over correctly ruling out negative cases, or vice versa.

3.11. Feature Ablation Study

A leave-one-feature-out ablation study was conducted on the top-performing soft voting ensemble framework to quantify each individual variable's structural contribution to the classification loop. As shown in Table 10, the cross-validated baseline performance utilizing the entire feature set yields an baseline accuracy of 95.76% and an F_1-score of 95.79%.[1]

Removing *cp* (chest pain type) produced the largest degradation in both accuracy (+1.71 percentage points drop) and F_1-score (+1.73 percentage points drop), followed by *thal* (+1.19 points) and *restecg* (+0.57 points). Conversely, removing *chol* or *oldpeak* slightly improved accuracy (−0.47 and −0.41 points respectively), and removing *slope*, *exang*, or *thalach* had a negligible or marginally positive effect on performance. This behavior highlights feature redundancy within high-capacity, non-linear models; when a prominent feature is excluded, the ensemble architecture is capable of rerouting diagnostic information through highly correlated proxy parameters without significantly harming macro-level metrics. Figure 13

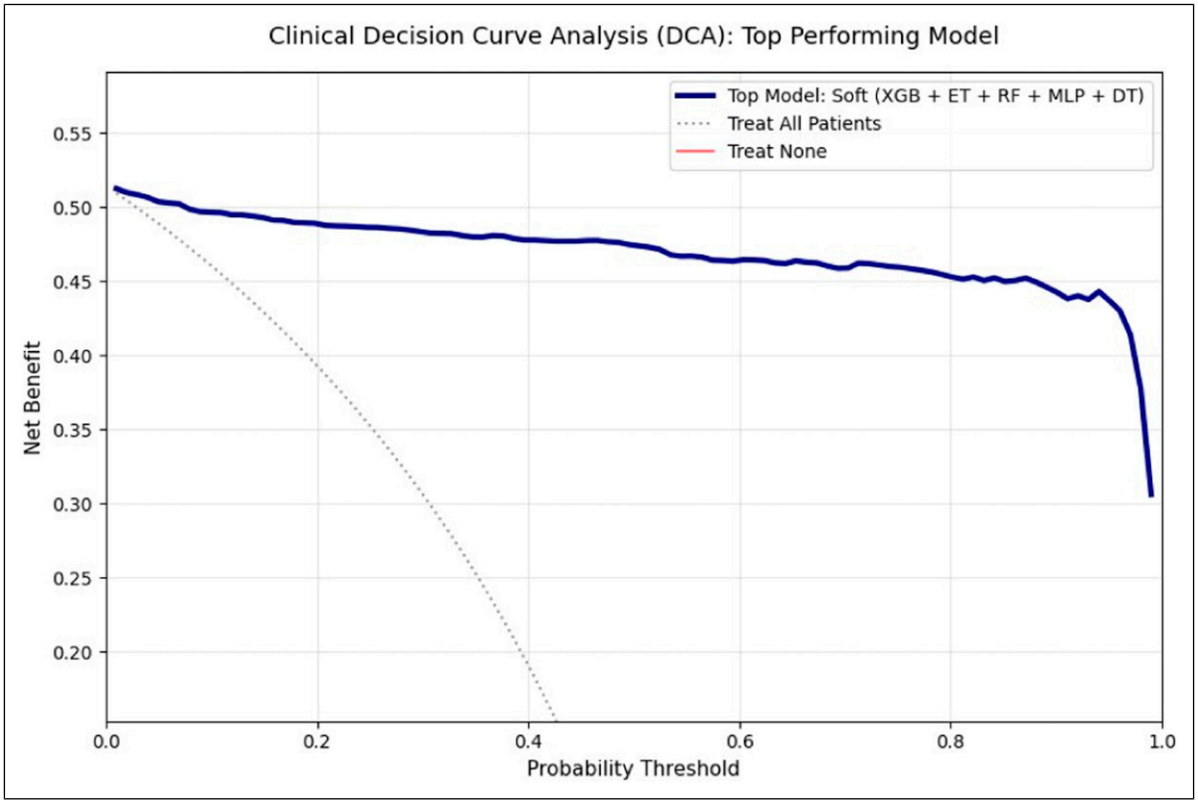


Figure 11. Clinical decision curve analysis comparing net benefit trends of the predictive ensemble against baseline target deployment models across varying threshold risk probabilities.

presents these results as a ranked bar chart, showcasing *cp*, *thal*, and *restecg* as the primary variables whose exclusion directly compromises accuracy.

4. Discussion

4.1. Interpretation of Overall Performance

The proposed framework achieved a top accuracy of 96.02% (soft-voting ensemble of XGBoost, Extra Trees, Random Forest, MLP, and Decision Tree), with a corresponding precision of 97.83% and recall of 94.37%. Unlike the inflated values obtained prior to correcting the evaluation pipeline, this result reflects performance under strict fold-wise separation of preprocessing, class balancing, and hyperparameter tuning, and should be considered the study's credible operating point.

Several observations follow from the results in Section 3. First, the gap between the best individual classifier (XGBoost, 95.86%) and the best ensemble (96.02%) is modest (0.16 percentage points), indicating that voting-based aggregation provided a real but incremental improvement rather than the primary source of predictive power, most of the discriminative signal was already captured by the strongest tree-based models individually. Second, tree-ensemble methods (XGBoost, Extra Trees, Random Forest) and MLP consistently outperformed linear and Naive Bayes models by 15–20 percentage points, suggesting that the relationship between the clinical features used and heart disease status in this dataset is substantially non-linear, consistent with known physiological interactions between variables such as chest pain type, ST-segment behavior, and vessel involvement. Third, hyperparameter tuning produced only marginal gains over untuned defaults (Section 3.2), which suggests the feature space and classifier defaults were already reasonably well-matched to this task, rather than indicating a shortcoming in the tuning procedure itself.

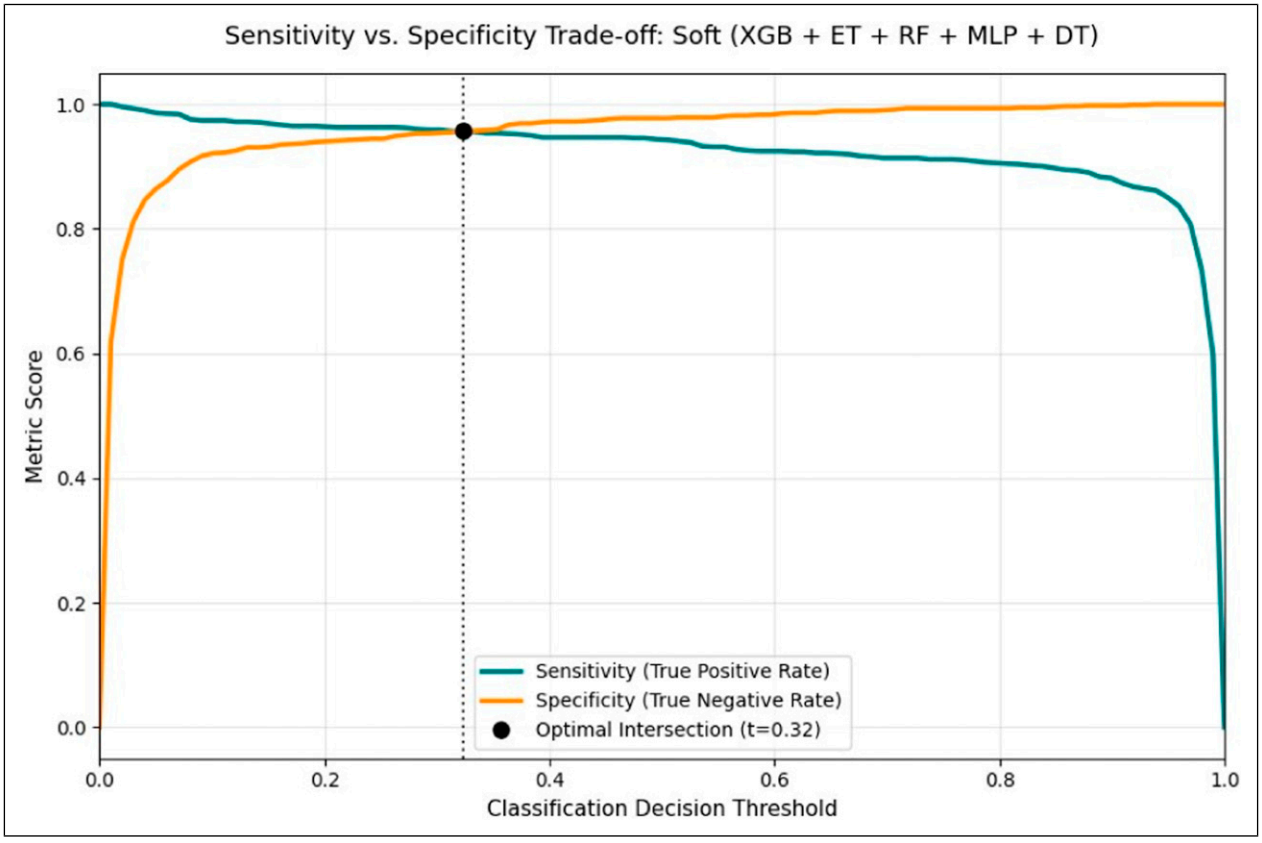


Figure 12. Sensitivity-specificity operational trade-off curve mapping parameter variation across classification cutoff values to locate intersection and balance parameters.

4.2. Comparison with Prior Studies

Table 11 situates the present findings within the context of prior heart disease prediction studies. Unlike several earlier works, the present study reports accuracy under a fold-wise, leakage-controlled validation protocol with nested hyperparameter tuning, rather than a single train-test split.

The proposed model's accuracy falls within the same range as the strongest prior results (e.g., Saboor et al.,[11] Li et al.[14]), rather than exceeding all prior work by a wide margin, which is itself informative: it suggests the present result reflects a genuinely competitive but realistic ceiling for this feature set and data source, rather than an artifact of an overly favorable

Table 10. Feature Ablation Study Results (Top Model: Soft XGB+ET+RF+MLP+DT).

Dropped Feature	Accuracy	F_1-Score	Accuracy Drop	F_1 Drop
None (Baseline, all features)	95.76%	95.79%	—	—
cp	94.05%	94.06%	+1.71%	+1.73%
thal	94.57%	94.61%	+1.19%	+1.18%
restecg	95.19%	95.20%	+0.57%	+0.59%
ca	95.45%	95.48%	+0.31%	+0.31%
age	95.55%	95.59%	+0.21%	+0.20%
trestbps	95.66%	95.69%	+0.10%	+0.10%
sex	95.66%	95.69%	+0.10%	+0.10%
slope	95.76%	95.81%	$\approx$ 0.00%	−0.02 %
exang	95.81%	95.86%	−0.05 %	−0.07 %
thalach	95.92%	95.94%	−0.15 %	−0.15 %
oldpeak	96.17%	96.21%	−0.41 %	−0.42 %
chol	96.23%	96.26%	−0.47 %	−0.47 %

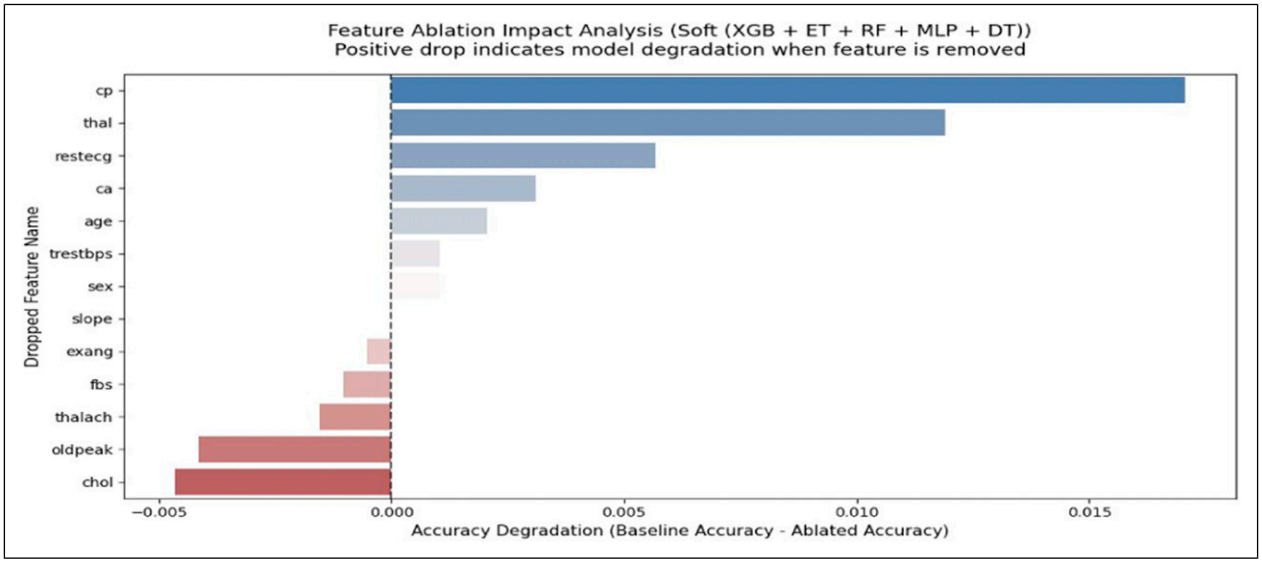


Figure 13. Ranked bar chart from the feature ablation process displaying measured shifts in validation performance metrics following isolated variable exclusion rounds.

evaluation protocol. Notably, most of the compared studies do not report cross-validated confidence intervals, calibration, or decision-curve analysis, and several rely on a single dataset without disclosing whether preprocessing was performed before or after data splitting, limitations the present study explicitly avoids or addresses (Sections 3.1–3.2, 3.8–3.9). Furthermore, while the majority of historical models operate as black boxes, the proposed framework explicitly embeds Explainable AI (XAI) to ensure clinical transparency.

4.3. Clinical Utility, Calibration, and Decision Curve Findings

In addition to raw discrimination, calibration and decision-curve analysis were also conducted to determine if outputs of the model are usable in a clinical decision-support context beyond accuracy. The calibration curve and the Brier score (Section 3.8) suggest that the model's predictions of probabilities are reliable, with the exception of a small range in the middle of the curve that is not consistent with a systematic error, because an accurate but poorly calibrated model can still lead to misleading predictions for the person interpreting the model's output as a probability.

The decision curve analysis (Section 3.9) shows that the model has a greater net benefit than "treat-all" and "treat-none" for nearly all of the range of the threshold. The benefit of the model diminishes with higher thresholds, beyond the threshold where "treat-all" is not helpful. The accuracy tables do not give this information: whether such use of the model would enhance the treatment decision, as it would be more representative of what is meant by clinically useful.

The sensitivity–specificity analysis (Section 3.10) indicated that an optimal threshold of 0.323 is much lower than the default threshold of 0.5. This lower threshold (compromise in specificity versus sensitivity) may be the more clinically appropriate operating point in a screening situation than the default threshold in the main classification tables, as a missed case of heart disease is likely to be a more costly false alarm.

Table 11. Comparison with Previous Studies on Heart Disease Prediction.

Author	Dataset	Method	XAI	Accuracy
Saboor A. et al.[11]	Cleveland Heart Disease	SVM	No	96.72%
Mohapatra et al.[15]	UCI Heart Disease	Stacking Ensemble	No	92.00%
Kavitha et al.[16]	Cleveland Heart Disease	Hybrid Model	No	88.70%
Shah, D. et al.[13]	Cleveland Heart Disease	KNN	No	90.79%
Li et al.[14]	Kaggle Heart Disease	EPSOSA-BP	Yes	95.20%
Proposed Model	**Kaggle + UCI + Cleveland**	**Soft Voting (XGB+ET+RF+MLP+DT)**	**Yes**	**96.02%**

4.4. Feature Importance and Clinical Plausibility

Consistent with the previous results from the feature ablation study (Section 3.11), both the results of SHAP (Section 3.7) indicated that chest pain type and thalassemia status were the features most important for the prediction of the disease; higher severity chest pain and abnormal thal values both skewed the prediction toward disease, which aligns with the direction of effect that is common in the cardiology field, and not an artifact of the dataset. One thing to note is that ST-segment slope is ranked as a high feature by the per-prediction spread, but removing it didn't affect the overall accuracy much as ablation showed, possibly due to its high correlation with other features like oldpeak, but not really a contradiction and a good reminder that per-prediction attribution and aggregate feature necessity are not the same.

Small negative values for the ablation of cholesterol and oldpeak indicated that the removal of these variables slightly enhanced the accuracy. This does not indicate a lack of clinical relevance but is more likely to be noise or redundancy in the merged dataset for this particular feature.

4.5. Limitations

The clearest limitation is that the three merged datasets contain known record-level overlap that was not resolved through deduplication in this study. Fold-wise cross-validation prevents leakage from preprocessing performed before splitting, but it does not prevent duplicate patient records from landing in both train and test partitions of the same fold. Reported accuracy should therefore be read as an upper-bound estimate rather than a guaranteed one.

Beyond this, no external validation dataset was used, so generalizability outside this specific data pool remains unconfirmed; the merged sample (n = 1,934) and 13-variable feature set are modest relative to what richer clinical data (imaging, biomarkers, longitudinal records) could offer; SHAP was applied to XGBoost as the ensemble's leading constituent rather than to the ensemble jointly, since exact multi-model attribution isn't directly supported by standard SHAP implementations; and no comparison was made against established clinical risk scores such as Framingham or ASCVD/SCORE2. Resolving dataset overlap, validating externally, benchmarking against these risk scores, and extending toward multi-modal data are the clearest next steps for this work.

5. Conclusion

In this study, we developed a machine learning framework for heart disease prediction that combines hybrid data balancing, ensemble learning, and SHAP-based interpretability, evaluated under a leakage-controlled, fold-wise validation protocol with nested hyperparameter tuning. Three publicly available datasets were combined into a single pooled source of 1,934 records to improve model generalization, and a hybrid balancing technique combining Instance Hardness Threshold (IHT) and Tomek Links was applied strictly within each training fold to address class imbalance and noisy class boundaries without leaking information into held-out test data.

A soft-voting ensemble of classifiers (XGBoost, Extra Trees, Random Forest, MLP, and Decision Tree) outperformed the other classifiers, with accuracy of 96.02% (95% CI: 95.47–96.56%), precision of 97.83%, and recall of 94.37%. In addition to classification accuracy, the model was also evaluated for its practical clinical utility through calibration analysis and clinical decision-curve analysis, along with a sensitivity–specificity threshold analysis with an optimal threshold of 0.323, a sensitivity of 95.68%, and a specificity of 95.74%. The SHAP-based interpretability and a complementary feature ablation study showed that the most influential features were chest pain type, thalassemia status, and results of the resting electrocardiogram, which are all known cardiovascular risk factors, and supported the clinical plausibility of the model's decision-making.

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Ethical considerations

This study was conducted exclusively using existing, publicly available, de-identified secondary datasets obtained from publicly accessible sources. The authors did not conduct any interventions, prospectively recruit individuals, or obtain personally identifiable information. Thus, an ethics approval number and permission from an Institutional Review Board (IRB) or Institutional Ethics Committee (IEC) were not necessary for this investigation. The datasets were ethically sourced from public repositories, and the work was conducted in accordance with the principles of the Declaration of Helsinki, as applicable to research utilizing existing anonymized data. Informed consent was not required because no direct participant involvement occurred.

Consent for publication

Not applicable – this manuscript does not contain any individual patient data, images, or identifiable information requiring consent for publication.

Author contributions

Khandaker Mohammad Mohi Uddin: Conceptualization, Data Curation, Investigation, Writing – Original Draft, Visualization, Methodology; Abdullah: Formal Analysis, Resources, Writing – Original Draft; Chowdhury Mohammad Rakin Haider: Methodology, Validation, Writing – Review & Editing, Supervision.

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Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

The datasets analyzed in this study are publicly available and required no special permission for use: the Heart Disease Dataset from Kaggle,[22] the UCI Heart Disease Dataset,[23] and the Heart Disease Cleveland Dataset.[24] The merged dataset and analysis code generated for this study are available from the corresponding author upon reasonable request.

Guarantor

Khandaker Mohammad Mohi Uddin serves as the data guarantor for this study.

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